

Lasso and coordinate descent

Swap the L2 penalty for L1: corners that set coefficients to exactly 0, subgradients at the kink, soft thresholding one coordinate at a time, the lasso path and feature selection.

THE QUESTION

Ridge shrinks every coefficient but keeps them all. How do we fit well and keep only the features that matter?

THE INTUITION

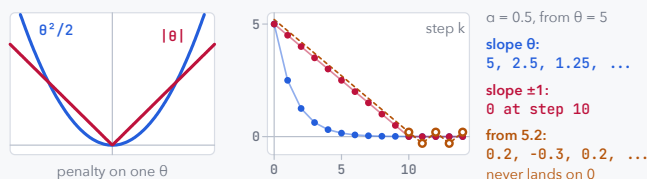
Penalise $\lambda \sum_j |\theta_j|$ instead of $\lambda \sum_j \theta_j^2$. The L1 ball has corners on the axes, so the loss contours tend to touch it where some $\theta_j = 0$. Solving for one θ_j at a time gets round the kink at 0 and returns exactly 0 for a weak feature.

KEY TAKEAWAY

Lasso minimises $\text{SSE} + \lambda \|\theta\|_1$. Coordinate descent repeats $\theta_j = S(\rho_j, \lambda/2)/z_j$, which is 0 whenever $|\rho_j| \leq \lambda/2$. Larger λ zeroes more coefficients; pick λ by cross-validation.

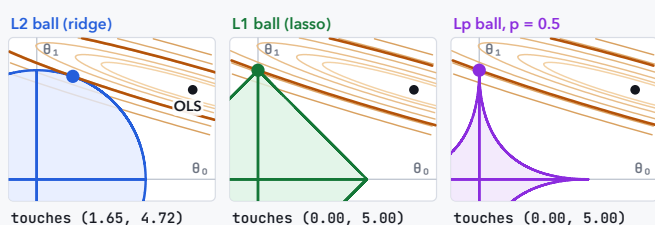
ridge keeps every $\theta_j \rightarrow$ add $\lambda \sum |\theta_j| \rightarrow$ kink at 0: subgradient $\rightarrow$ update one θ_j at a time $\rightarrow$ soft threshold gives exact 0s

1 Lasso charges $|\theta_j|$ instead of θ_j^2



Lasso (least absolute shrinkage and selection operator) minimises $\|y - X\theta\|^2 + \lambda \|\theta\|_1$ with $\|\theta\|_1 = \sum_j |\theta_j|$, or the SSE subject to $\|\theta\|_1 \leq s$; each s matches a λ . The slope of $\theta^2/2$ is θ , so its pull fades near 0; the slope of $|\theta|$ stays ± 1 . With step 0.5 from 5, L2 halves forever and L1 hits 0 at step 10, only because 5 is a multiple of 0.5: from 5.2 it hops between 0.2 and -0.3 .

2 Corners on the axes give exact zeros

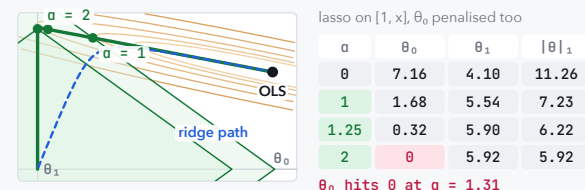


One loss (the line of panel 4), three balls of size 5. The tightest contour meets the L2 ball off the axes at (1.65, 4.72) but the L1 diamond at its corner (0, 5), so $\theta_0 = 0$. For $p < 1$ the corners are sharper and zeros more common, but the ball is not convex and the fit can stall in a local minimum. $p = 1$ is the smallest p whose ball is convex.

3 Lasso or ridge?

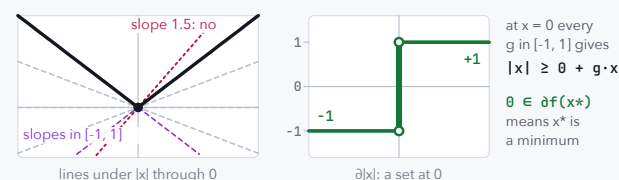
	Ridge (L2)	Lasso (L1)
Penalty	$\lambda \sum_j \theta_j^2$	$\lambda \sum_j \theta_j $
Ball	circle	diamond
Exact zeros	never	yes, so it selects
At $\theta_j = 0$	smooth	a kink
Solved by	closed form	coordinate descent
Correlated Suits	shares the weight many small effects	keeps one few that matter

4 Raising α walks θ onto an axis



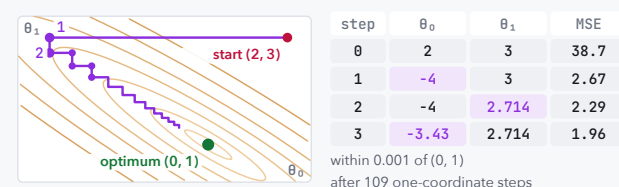
The lasso notebook's 60 points, $y = 4x + 7$ plus noise ($\sigma = 3$), columns 1, x , loss $\frac{1}{2N} \text{SSE} + \alpha \|\theta\|_1$ (so $\lambda = 2N\alpha$). At $\alpha = 1$ the fit is (1.68, 5.54), $\|\theta\|_1 = 7.23$, as on the slides. θ_0 hits 0 at $\alpha = 1.31$ while θ_1 rises to 5.98, then the path runs down the axis. Ridge never lands on an axis. The intercept is penalised here only for the picture.

5 A subgradient is a slope that stays under f



g is a subgradient of a convex f at x_0 if $f(x) \geq f(x_0) + g(x - x_0)$ for all x ; where f is differentiable, $f'(x_0)$ is the only one. $\partial|x|$ is -1 for $x < 0$, $[-1, 1]$ at 0, $+1$ for $x > 0$. x^* minimises a convex f exactly when $0 \in \partial f(x^*)$.

6 Coordinate descent solves for one θ_j at a time



Minimise over one coordinate with the rest fixed and cycle through j until nothing moves; no step size. The slides' example: the three points of the gradient descent sheet from (2, 3). With $\theta_1 = 3$, $(6\theta_0 + 24)/3 = 0$ gives $\theta_0 = -4$; then $\theta_1 = 2.714$ and $\theta_0 = -3.43$ (slides: 2.7, -3.4). The $\theta_0\theta_1$ term couples them, so the path zigzags; the separable $(\theta_0 - 2)^2 + (\theta_1 - 1)^2$ needs one sweep.

Soft thresholding sets weak coefficients to 0

The lasso coordinate update, when coordinate descent stalls, the lasso path on real data, choosing λ , and stepwise selection.

7 The lasso update is a soft threshold



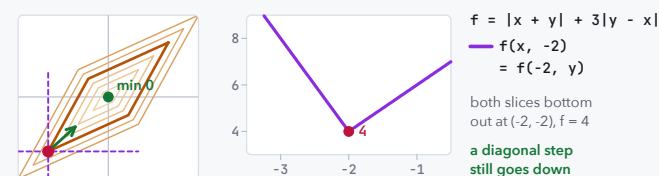
Fix all but θ_j . With $\rho_j = \sum_i x_{ij}(y_i - \hat{y}_i^{(-j)})$, the residual without feature j against x_j , and $z_j = \sum_i x_{ij}^2$: $\partial \text{SSE} / \partial \theta_j = -2\rho_j + 2\theta_j z_j$, so least squares gives ρ_j / z_j . The lasso needs $0 \in -2\rho_j + 2\theta_j z_j + \lambda \partial |\theta_j|$:

$$\theta_j = \begin{cases} (\rho_j + \lambda/2) / z_j & \text{if } \rho_j < -\lambda/2 \\ 0 & \text{if } |\rho_j| \leq \lambda/2 \\ (\rho_j - \lambda/2) / z_j & \text{if } \rho_j > \lambda/2 \end{cases}$$

or $S(\rho_j, \lambda/2) / z_j$ with $S(\rho, t) = \text{sign}(\rho) \max(|\rho| - t, 0)$. A weak correlation gives exactly 0; the ridge update $\rho_j / (z_j + \lambda)$ only rescales. In NumPy (checked against our solver):

```
def lasso_cd(X, y, lam, sweeps=200):
    w = np.zeros(X.shape[1])
    z = (X ** 2).sum(axis=0)
    for _ in range(sweeps):
        for j in range(len(w)):
            w[j] = 0.0 # drop j
            rho = X[:, j] @ (y - X @ w)
            t = max(abs(rho) - lam / 2, 0)
            w[j] = np.sign(rho) * t / z[j]
    return w
```

8 Coordinate descent can stall at a coupled kink



The slides' $f = |x + y| + 3|y - x|$ from $(-2, -2)$, where $f = 4$. Moving x or y alone raises f both ways, so coordinate descent stops, yet the minimum is 0 at the origin. The lasso is safe: its kinked part $\sum_j |\theta_j|$ is one term per coordinate and the SSE is smooth, and then cyclic coordinate descent converges.

12 Pause and think

Gradient descent? Hops around 0.
Elastic net? L1 + L2 keeps groups.

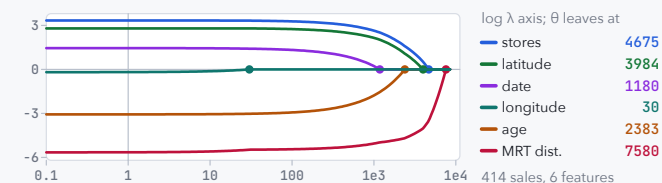
All zero when? $\lambda \geq 2 \max_j |x_j^T y|$.
Step size? None: exact 1-D solves.

Why $p = 1$? Corners, yet convex.
Before fitting? Standardise; free θ_0 .

minimise $\text{SSE} + \lambda \|\theta\|_1$ · corners give exact zeros · $\theta_j = S(\rho_j, \lambda/2) / z_j$ · bigger λ , fewer features · choose λ by CV

Before: the ridge regression sheet; next: the logistic regression sheet. Source: lecture slides "Lasso regression", "Coordinate descent", "Subgradient" and "Some practical tips and feature selection", and the lasso notebook with its 60-point line and real-estate table (nipunbatra.github.io/ml-teaching). The 40-feature data and CV shuffles are ours, seeded. Corrected: the negative case of the slides' update ($\rho_j < -\lambda/2$), the stepwise count $d(d+1)/2$, and the touching point in the L1 sketch.

9 On real data coefficients reach exactly 0



The ridge sheet's real-estate table, six standardised features. Each coefficient hits 0 at its own λ (dots), longitude first; all are 0 beyond $\lambda_{\max} = 2 \max_j |x_j^T y| = 7580$. Ridge kept all six even at $\lambda = 10^6$. The slides' plot (unit-norm columns) has the same order.

10 Cross-validation picks λ ; lasso finds the few that matter



Our seeded data: 60 rows, 40 features, 5 that matter. 5-fold CV picks $\lambda = 7.9$, which keeps 21 features including all 5; CV aims at prediction, so noise features keep small weights. Test MSE: lasso 1.49, ridge 2.27, least squares 3.16.

11 Stepwise selection searches subsets greedily



Instead of all 2^d subsets, forward selection adds the feature that lowers the validation loss most, round by round; backward drops the one whose removal hurts least. Each takes $d + (d-1) + \dots + 1 = d(d+1)/2$ fits. On the real-estate table longitude, added last, makes the CV error worse.